



Solutions to First Order DE

Continuation ...

3. Exact type

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Standard form:

$$M(x,y)dx + N(x,y)dy = 0$$

General solution:

$$F(x,y) = C$$

$$\text{where: } F(x,y) = \int M(x,y)dx + k_1$$

$$F(x,y) = \int N(x,y)dy + k_2$$

Test for exactness:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

4. Linear type

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If a differential equation is not exact, an introduction of a certain factor called the integrating factor makes it exact. This can be demonstrated by linear equations of order one.

Below is the standard form of an equation that is linear and of order one in the dependent variable y.

Standard form:

$$\frac{dy}{dx} + yP(x) = Q(x)$$

General solution:

$$y(\text{i.f.}) = \int Q(x)(\text{i.f.})dx + C$$

where: i.f. = integrating factor

$$\text{i.f.} = e^{\int P(x)dx}$$

Note: Integrating factor is also known as Euler's multiplier

5. Bernoulli's equation type

Standard form:

$$\frac{dy}{dx} + yP(x) = y^n Q(x)$$

In this type of DE, n is not equal to 1. Otherwise, it becomes a separable equation.

General solution:

$$y^{1-n} = \frac{1}{e^z} \int (1-n)Q(x)e^z dx$$

where:

$$z = \int (1-n)P(x)dx$$

The Bernoulli Equation is named after the brothers Jakob (1654 - 1705) and Johann Bernoulli (1667 - 1748).

Applications of First Order Differential Equations

There are many occasions in nature that first order differential equations arise. Hence, differential equations can be used in solving these problems.

1. Population growth problems / Population dynamics

$$\frac{dP}{dt} = kP$$

where:

$$\frac{dP}{dt} = \text{rate of change of the population}$$

P = no. of inhabitants at any time t

k = constant of proportionality

2. Radioactive decay problems

Radioactive substances are those that naturally and spontaneously break down into their elements. As a result, they release energy once they break down. The rate of change of the substance (or rate of decay) is proportional to the amount of the substance that is present.

$$\frac{dQ}{dt} = kQ$$

where:

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$$\frac{dQ}{dt} = \text{rate of change of the substance}$$

Q = amt of substance at any time t

k = constant of proportionality

3. Economics: Continuous compound interest problems

$$\frac{dP}{dt} = rP$$

where:

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$$\frac{dP}{dt} = \text{rate of change of the account}$$

P = money present in the account at any time t

r = nominal rate of interest per year

4. Temperature: Cooling and heating problems

$$\text{Cooling process } \frac{dT}{dt} = k(T - t_s)$$

$$\text{Heating process } \frac{dT}{dt} = k(t_s - T)$$

where:

$$\frac{dT}{dt} = \text{rate of change of the body temp}$$

T = temp of body at any time t

t_s = surrounding temp

5. Flow problems & Mixture problems

$$\frac{dQ}{dt} = \text{rate of gain} - \text{rate of loss}$$

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where:

Q = concentration of the mixture at any time t

$$\frac{dQ}{dt} = \text{rate of change of the concentration of the mixture}$$

6. Kinematics and Newton's second law of motion

Differential Equations can be used to solve problems in Kinematics where displacement, velocity and acceleration are the dependent variables and time is the independent variable.

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

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